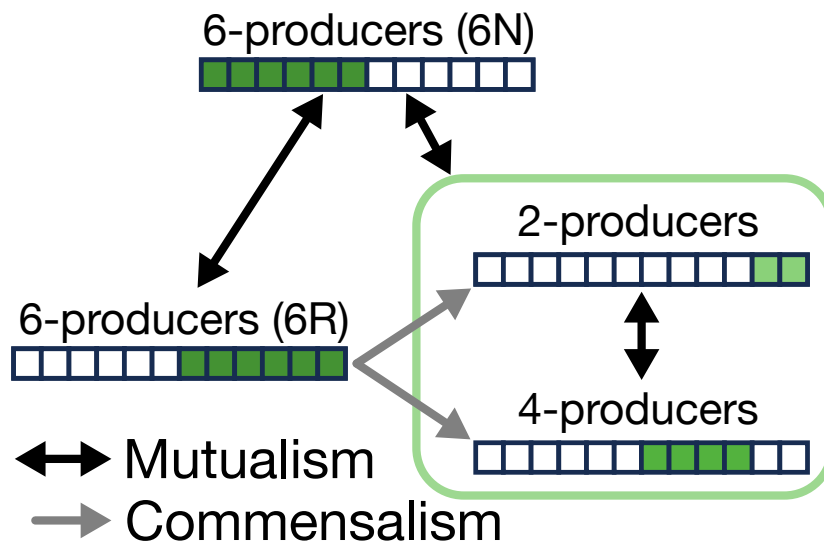
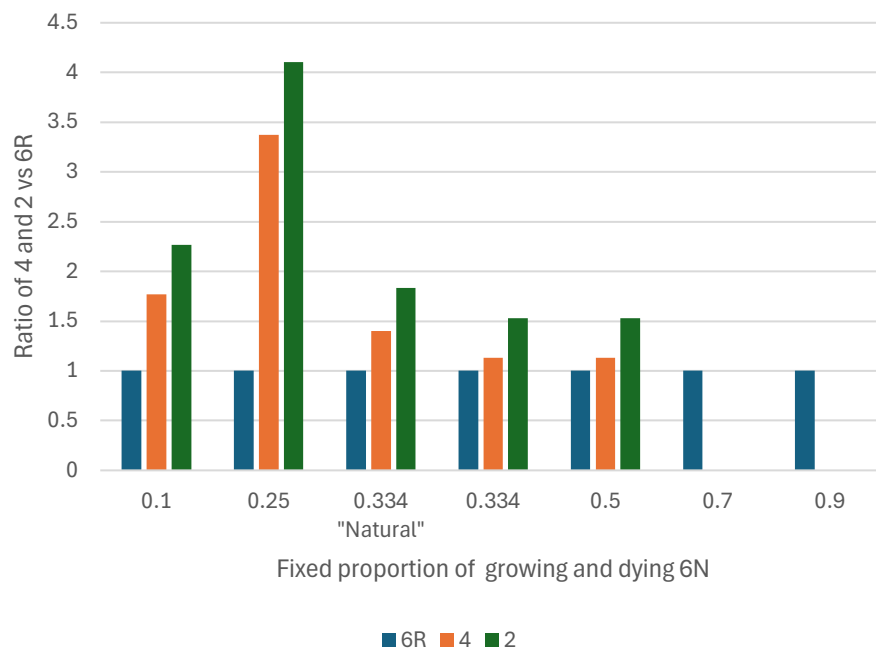


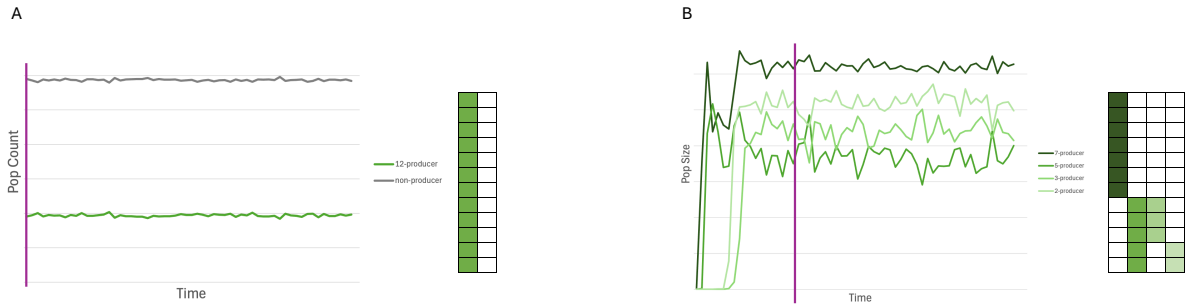




**Figure S3.** Interactions between the ecotypes in the 6-(6 OR 4-2) system.



**Figure S4.** 6N as living blocks. A modification of the 6-(6 OR 4-2) system. 6N is altered to only grow or die when below or above, respectively, to maintain a fixed proportion of 6N cells. 6R, 4, and 2 are initialised at their equilibrium proportions to each other using the amount of free space after initialising 6N. Bars show the ratios of 4 and 2 against 6R, to allow for changing absolute numbers with different numbers of 6N.



**Figure S5.** Simulations with super deletions. Simulations started with 12 common goods, a CG cost of 1E-02, default mutation rate and no HGT. A special super deletion parameter was added which allows all production to be converted to non-production in a single event (5.00E-09), i.e., the cell becomes a pure non-producer. **A)** Example of an early superdeletion event. Non-producers rapidly overtake the system, trapping it in a 12-0 configuration. **B)** Example of a late superdeletion event. The system had already formed a quasi-stable 7-(5 OR 3-2) configuration and the non-producer was unable to invade.

| Default Parameters                                                           | Default values                                                                                                             |
|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Grid Size                                                                    | 200x200 cells (40,000 carrying capacity)                                                                                   |
| Baseline reproduction chance                                                 | 0.6                                                                                                                        |
| Baseline death rate per timestep                                             | 0.1                                                                                                                        |
| Number of common goods (CGs)                                                 | 6 or 12                                                                                                                    |
| Growth penalty per CG produced (Cost per CG * Baseline reproduction chance ) | 1e-04, 1e-03, 1e-02, 1e-01                                                                                                 |
| Instantaneous chance of reproduction                                         | Baseline reproduction chance (0.6) - (#CGs produced * Growth penalty per CG produced)                                      |
| Interaction range (Moore neighborhood radius)                                | 1 / 2 / 3 / 4 (radius of moore neighbourhood) ;<br>9 / 25 / 49 / 81 (number of potential beneficiaries, producer included) |
| Mutation rate / per gene / per timestep                                      | 5e-06                                                                                                                      |
| Horizontal Gene Transfer rate                                                | 0                                                                                                                          |

**Table S1.** Default model parameters.

| Run # | % Invader | Timesteps | Invader ecotypes | Defender ecotypes | Surviving ecotypes | Lost ecotypes | Ecotypes on way out |
|-------|-----------|-----------|------------------|-------------------|--------------------|---------------|---------------------|
| 1     | 10        | 30k       | 6633             | 6642              | 6642               | 33            | N/A                 |
| 2     | 10        | 30k       | 6633             | 6642              | 6642               | 33            | N/A                 |
| 3     | 10        | 30k       | 6633             | 6642              | 6642               | 33            | N/A                 |
| 1     | 10        | 30k       | 6642             | 6633              | 6633               | 42            | N/A                 |
| 2     | 10        | 30k       | 6642             | 6633              | 6633               | 42            | N/A                 |

|   |    |     |      |      |        |     |     |
|---|----|-----|------|------|--------|-----|-----|
| 3 | 10 | 30k | 6642 | 6633 | 6633   | 42  | N/A |
| 1 | 50 | 20k | 6633 | 6642 | 664233 | N/A | 33  |
| 2 | 50 | 20k | 6633 | 6642 | 664233 | N/A | 3   |
| 3 | 50 | 20k | 6633 | 6642 | 6642   | 33  | N/A |
| 4 | 50 | 20k | 6633 | 6642 | 664233 | N/A | 3   |
| 5 | 50 | 20k | 6633 | 6642 | 664233 | N/A | 33  |
| 1 | 50 | 20k | 6642 | 6633 | 6642   | 33  | N/A |
| 2 | 50 | 20k | 6642 | 6633 | 66423  | 3   | 3   |
| 3 | 50 | 20k | 6642 | 6633 | 6642   | 33  | N/A |
| 4 | 50 | 20k | 6642 | 6633 | 664233 | N/A | 33  |
| 5 | 50 | 20k | 6642 | 6633 | 66423  | 3   | 3   |

**Table S2.** 6-(6 or 4-2) and 6-(6 or 3-3) invasions results. Invasions used 10% invaders and 50% invaders. See Supplementary Data for raw data.